

MH3100 Real Analysis I

Tutorial 5 Solutions

Academic Year 2026/2027, Semester 1

Quantitative Research Society @NTU

October 1, 2026

Overview

- **Lecture coverage:** Lectures 13–17 — limit points, isolated points and closures (L13–14); closed sets (L15); extrema of closed and bounded sets and (sequential) compactness (L16); the Nested Set Property (L17).
- **Questions:** 8 questions (5 main + 3 Additional) from Tutorial 5.
- **Notation:** $\mathbb{N} = \{1, 2, 3, \dots\}$ (0 is not a natural number in this course); closure of A is $\overline{A} = \overline{A}$. **This course never uses open sets, open covers, or the Heine–Borel Theorem:** compactness is always (*sequentially*) *compact* in the sense of the course’s own definition (every sequence in the set has a subsequence converging to a limit in the set).

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Final Answers

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Question 1 (Limit points, isolated points, closures of five sets)

Problem

For each of the following sets, determine (i) the set of all its limit points; (ii) the set of all its isolated points; and (iii) its closure.

(a) $A = \left\{ (-1)^n + \frac{2}{n} : n \in \mathbb{N} \right\}.$

(b) $B = \{x \in \mathbb{Q} : 0 < x < 1\}.$

(c) $C = \{x \in \mathbb{R} : x \neq 0\}.$

(d) $D = \left\{ 1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} : n \in \mathbb{N} \right\}.$

(e) $E = \left\{ 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} : n \in \mathbb{N} \right\}.$

(Tutorial 5, Question 1; Lectures 13–15, limit points / isolated points / closure.)

Solution

Method 1: Sequential Criterion for Limit Points

Lemma. Let $\{a_n\}$ be injective, $C := \{a_n : n \in \mathbb{N}\}$, $L(C) :=$ the limit points of C . Then $x \in L(C) \iff$ some subsequence of $\{a_n\}$ has limit x . Isolated points $= C \setminus L(C)$ and $\overline{C} = C \cup L(C)$.

Proof of Lemma. $(\Rightarrow)$ There is $\{y_j\}$ in $C \setminus \{x\}$ with $x = \lim y_j$ (Sequential Criterion for Limit Points). Write $y_j = a_{k_j}$. If $\{k_j\}$ were bounded, $\{y_j\}$ would take values in a finite set $F \subseteq C \setminus \{x\}$. Then $\delta := \min_{f \in F} |f - x| > 0$ and $|y_j - x| \geq \delta$ for all j , contradicting $x = \lim y_j$. $\therefore \{k_j\}$ is unbounded; choose $j_1 < j_2 < \cdots$ with $k_{j_1} < k_{j_2} < \cdots$. $\{a_{k_{j_i}}\}$ is a subsequence of $\{a_n\}$ and of $\{y_j\}$. $\therefore x = \lim a_{k_{j_i}}$ (Subsequential Limits).

$(\Leftarrow)$ Let $x = \lim a_{n_i}$. Injectivity gives $a_{n_i} = x$ for at most one i . $\therefore$ a tail of $\{a_{n_i}\}$ lies in $C \setminus \{x\}$ and has limit x , so $x \in L(C)$ (Sequential Criterion for Limit Points). $\square$

(a) $A = \{(-1)^n + 2/n : n \in \mathbb{N}\}.$

Proof. $a_n := (-1)^n + \frac{2}{n}$ is injective: odd terms $-1 + \frac{2}{n} \leq 1$ and even terms $1 + \frac{2}{n} > 1$, each strictly decreasing in n . Let $x \in L(A)$; by the Lemma $x = \lim a_{n_i}$ for a subsequence. Infinitely many n_i have one parity; along them $a_{n_i} = \pm 1 + \frac{2}{n_i}$ has limit ± 1 ($n_i \geq i$). $\therefore x = \pm 1$ (Subsequential Limits). Conversely $-1 = \lim a_{2k-1}$ and $1 = \lim a_{2k} \therefore L(A) = \{-1, 1\}$ (Lemma). $1 = a_1 \in A$; $-1 \notin A$ (odd terms > -1 , even terms > 1). $\therefore$ isolated points $= A \setminus \{1\}$ and $\overline{A} = A \cup \{-1\}$. $\square$

(a) Limit points $\{-1, 1\}$; isolated points $A \setminus \{1\}$; $\overline{A} = A \cup \{-1\}$.

Density Lemma. If $0 \leq a < b$ then $a < q < b$ for some $q \in \mathbb{Q}$.

Proof of Density Lemma. Choose $m \in \mathbb{N}$ with $m(b-a) > 1$ (Archimedean). $\{j \in \mathbb{N} : j > ma\} \neq \emptyset$ (Archimedean), so it has a least element k (well-ordering). $k-1 \leq ma$ ($k=1$: $ma \geq 0$; $k \geq 2$: $k-1$ is not in the set) $\therefore ma < k \leq ma+1 < mb \therefore a < \frac{k}{m} < b$. $\square$

(b) $B = \mathbb{Q} \cap (0, 1)$.

Proof. Let $x \in [0, 1]$. By the Density Lemma choose rationals q_n with

$$q_n \in \left(x, \min\left\{x + \frac{1}{n}, 1\right\}\right) \quad (x < 1), \quad q_n \in \left(\max\left\{1 - \frac{1}{n}, 0\right\}, 1\right) \quad (x = 1).$$

Then $q_n \in (0, 1) \cap \mathbb{Q} = B$, $q_n \neq x$ and $|q_n - x| < \frac{1}{n} \therefore x = \lim q_n$ (Squeeze), so $x \in L(B)$. If $x < 0$ let $\varepsilon := -x$; if $x > 1$ let $\varepsilon := x - 1$. Then $(x - \varepsilon, x + \varepsilon) \cap (0, 1) = \emptyset$, so $x \notin L(B)$. $\therefore L(B) = [0, 1]$; as $B \subseteq [0, 1]$, isolated points $= B \setminus [0, 1] = \emptyset$ and $\overline{B} = [0, 1]$. $\square$

(b) Limit points $[0, 1]$; isolated points $\emptyset$; $\overline{B} = [0, 1]$.

(c) $C = \mathbb{R} \setminus \{0\}$.

Proof. Let $x \in \mathbb{R}$. $x + \frac{1}{n} = 0$ for at most one $n \in \mathbb{N}$, so there is N with $y_n := x + \frac{1}{n} \in C \setminus \{x\}$ for all $n \geq N$. $x = \lim y_n \therefore x \in L(C)$ (Sequential Criterion for Limit Points). $\therefore L(C) = \mathbb{R}$, isolated points $= C \setminus \mathbb{R} = \emptyset$, $\overline{C} = \mathbb{R}$. $\square$

(c) Limit points $\mathbb{R}$; isolated points $\emptyset$; $\overline{C} = \mathbb{R}$.

(d) $D = \{1 + \frac{1}{4} + \dots + \frac{1}{n^2} : n \in \mathbb{N}\}$.

Proof. $s_n := \sum_{k=1}^n \frac{1}{k^2}$ is strictly increasing, so injective, and

$$s_n \leq 1 + \sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k}\right) = 2 - \frac{1}{n} < 2 \quad \left(\frac{1}{k^2} < \frac{1}{k(k-1)}, k \geq 2\right)$$

$\therefore s = \sup s_n = \lim s_n$ exists (MCT), and every subsequence has limit s (Subsequential Limits). $\therefore L(D) = \{s\}$ (Lemma); $s \geq s_{n+1} > s_n$ for all $n \therefore s \notin D$. $\therefore$ isolated points $= D$ and $\overline{D} = D \cup \{s\}$. $\square$

(d) Limit points $\{s\}$, $s = \sum_{n=1}^{\infty} 1/n^2$; isolated points D ; $\overline{D} = D \cup \{s\}$.

(e) $E = \{1 + \frac{1}{2} + \dots + \frac{1}{n} : n \in \mathbb{N}\}$.

Proof. $t_n := \sum_{k=1}^n \frac{1}{k}$ is strictly increasing, so injective.

$$t_{2N} - t_N = \sum_{k=N+1}^{2N} \frac{1}{k} \geq N \cdot \frac{1}{2N} = \frac{1}{2} \quad \text{for every } N \in \mathbb{N}$$

$\therefore \{t_n\}$ is not Cauchy, so diverges (Cauchy Criterion); increasing, so unbounded above (MCT). A subsequence $\{t_{n_i}\}$ has $n_i \geq i$, so $t_{n_i} \geq t_i$: unbounded, hence divergent (convergent $\implies$ bounded). $\therefore L(E) = \emptyset$ (Lemma), isolated points $= E$, $\overline{E} = E$. $\square$

(e) Limit points $\emptyset$; isolated points E ; $\overline{E} = E$.

Set	Limit points	Isolated points	Closure
A	$\{-1, 1\}$	$A \setminus \{1\}$	$A \cup \{-1\}$
B	$[0, 1]$	$\emptyset$	$[0, 1]$
C	$\mathbb{R}$	$\emptyset$	$\mathbb{R}$
D	$\{s\}$	D	$D \cup \{s\}$
E	$\emptyset$	E	E

Question 2 ((Sequential) compactness of five sets)

Problem

Determine if each of the following sets is (sequentially) compact.

- (a) $A = [0, 1] \cup [5, 6]$.
- (b) $B = \mathbb{R}_0^+$.
- (c) $C = \{x \in \mathbb{R} : 0 \leq x \leq 1 \text{ and } x \text{ is irrational}\}$.
- (d) $D = \left\{1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} : n \in \mathbb{N}\right\}$.
- (e) $E = \left\{1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} : n \in \mathbb{N}\right\}$.

(Tutorial 5, Question 2; Lectures 16–17, Sequential Compactness / Extrema of Closed and Bounded Sets; builds on Question 1.)

Solution

Method 1: closed and bounded; witnessing sequences

- (a) $A = [0, 1] \cup [5, 6]$.

Proof. $A \subseteq [0, 6]$ is bounded. Let $a = \lim a_n$ with every $a_n \in A$. Infinitely many a_n lie in $[0, 1]$ or in $[5, 6]$; say $[0, 1]$ (other case identical). That subsequence has limit a (Subsequential Limits) and lies in $[0, 1]$ $\therefore 0 \leq a \leq 1$ (OLT), so $a \in A$. $\therefore A$ is closed (Sequential Criterion for Closed Sets); closed and bounded, so compact (Sequential Compactness of Closed and Bounded Sets). $\square$

(a) A is compact.

- (b) $B = \mathbb{R}_0^+$.

Proof. Let $b_n := n \in B$. A subsequence $\{b_{n_k}\}$ has $n_k \geq k$. For every $M > 0$ there is $k > M$ (Archimedean), so $b_{n_k} \geq k > M \therefore \{b_{n_k}\}$ is unbounded, hence divergent (convergent $\implies$ bounded). $\therefore \{b_n\}$ has no convergent subsequence. $\square$

(b) B is not compact.

- (c) $C = \{x \in [0, 1] : x \text{ irrational}\}$.

Proof. Let $x_n := \frac{\sqrt{2}}{n+2}$. $x_n = \frac{p}{q} \implies \sqrt{2} = \frac{p(n+2)}{q} \in \mathbb{Q}$, so $x_n \notin \mathbb{Q}$; $0 < x_n \leq \frac{\sqrt{2}}{3} < 1 \therefore x_n \in C$. $0 = \lim x_n$ and $0 \notin C$ ($0 \in \mathbb{Q}$); every subsequence has limit 0 (Subsequential Limits). $\therefore \{x_n\}$ has no subsequence with limit in C . $\square$

(c) C is not compact.

- (d) D .

Proof. By Question 1(d), $\{s_n\}$ lies in D with $s = \lim s_n \notin D$. Every subsequence has limit $s \notin D$ (Subsequential Limits) $\therefore \{s_n\}$ has no subsequence with limit in D . $\square$

(d) D is not compact.

(e) E .

Proof. By Question 1(e), $\{t_n\}$ lies in E and every subsequence is unbounded, hence divergent. $\therefore \{t_n\}$ has no convergent subsequence. $\square$

(e) E is not compact.

Set	(Sequentially) compact?
$A = [0, 1] \cup [5, 6]$	Yes
$B = \mathbb{R}_0^+$	No (unbounded)
C	No (not closed: irrationals $\rightarrow 0 \notin C$)
D	No (not closed: partial sums $\rightarrow s \notin D$)
E	No (unbounded)

Question 3 ($\sup A \in \overline{A}$ via the ε -Criterion for Closures)

Problem

Use the ε -Criterion for Closures to prove that if $A \subseteq \mathbb{R}$ is nonempty and bounded above, then $\sup A \in \overline{A}$. (*Tutorial 5, Question 3; Lecture 14, ε -Criterion for Closures; Lecture 1–3, ε -Criterion for Suprema.*)

Solution

Method 1: ε -Criterion for Suprema and for Closures

Proof. Let $s := \sup A$ (AoC) and $\varepsilon > 0$. $s - \varepsilon$ is not an upper bound of A (ε -Criterion for Suprema), so there is $x \in A$ with $s - \varepsilon < x$. $x \leq s \therefore |x - s| = s - x < \varepsilon$. Hence for every $\varepsilon > 0$ there is $x \in A$ with $|x - s| < \varepsilon$, so $s \in \overline{A}$ (ε -Criterion for Closures). $\square$

$\sup A \in \overline{A}$ for every nonempty, bounded-above $A \subseteq \mathbb{R}$.

Question 4 (Closed sets under union and arbitrary intersection)

Problem

Use the Sequential Criterion for Closed Sets to prove the following statements.

- (a) If F_1 and F_2 are closed sets, then so is their union $F_1 \cup F_2$.
- (b) If $\{F_t\}_{t \in T}$ is an arbitrary collection of closed sets, then so is their intersection

$$\bigcap_{t \in T} F_t := \{a : \forall t \in T \ a \in F_t\}.$$

(Tutorial 5, Question 4; Lecture 15, Sequential Criterion for Closed Sets; (a) reuses the “shuffled sequence” technique of Tutorial 3, Question 3.)

Solution

Method 1: Sequential Criterion for Closed Sets

(a) $F_1 \cup F_2$ is closed.

Proof. Let $c = \lim c_n$ with every $c_n \in F_1 \cup F_2$. $\mathbb{N} = \{n : c_n \in F_1\} \cup \{n : c_n \in F_2\}$ is infinite, so one of the two sets is infinite; say the first (other case identical). That subsequence lies in F_1 and has limit c (Subsequential Limits) $\therefore c \in F_1 \subseteq F_1 \cup F_2$ (F_1 closed). $\therefore F_1 \cup F_2$ is closed (Sequential Criterion for Closed Sets). $\square$

$$F_1, F_2 \text{ closed} \implies F_1 \cup F_2 \text{ closed.}$$

(b) $\bigcap_{t \in T} F_t$ is closed.

Proof. Let $a = \lim a_n$ with every $a_n \in \bigcap_{t \in T} F_t$. Let $t \in T$. $\{a_n\}$ lies in $F_t \therefore a \in F_t$ (F_t closed). t was arbitrary, so $a \in \bigcap_{t \in T} F_t \therefore$ the intersection is closed (Sequential Criterion for Closed Sets). $\square$

$$\{F_t\}_{t \in T} \text{ all closed} \implies \bigcap_{t \in T} F_t \text{ closed.}$$

Question 5 (A closed subset of a compact set is compact)

Problem

Use the definition of (sequential) compactness to prove that a closed subset of a (sequentially) compact set is (sequentially) compact; i.e., for any compact set $K \subseteq \mathbb{R}$ and any subset $A \subseteq K$, prove that A is compact if it is closed. **Do not use the Sequential Compactness of Closed and Bounded Sets or the Bolzano-Weierstrass Theorem.**

[Note: A proof without using the Sequential Compactness of Closed and Bounded Sets or the Bolzano-Weierstrass Theorem is applicable to compact sets in $\mathbb{Q}$ (or any ordered field) for which these properties do not hold.] (*Tutorial 5, Question 5; Lecture 16, definition of (sequential) compactness; Lecture 15, Sequential Criterion for Closed Sets.*)

Solution

Method 1: definition of compactness

Proof. Let K be compact and $A \subseteq K$ closed. Let $\{a_n\}$ be a sequence in A . $A \subseteq K \therefore \{a_n\}$ lies in K , so some subsequence $\{a_{n_k}\}$ has limit $x \in K$ (def. of compactness). $\{a_{n_k}\}$ lies in A and A is closed $\therefore x \in A$ (Sequential Criterion for Closed Sets). $\therefore$ every sequence in A has a subsequence with limit in A , i.e. A is compact. $\square$

$$\boxed{K \text{ compact and } A \subseteq K \text{ closed} \implies A \text{ compact.}}$$

Question A1 (Closed sets via distance-zero points)

Problem

For a nonempty set $A \subseteq \mathbb{R}$ and a number $a \in \mathbb{R}$, we define the *distance of a from A* as the infimum of the set $\{|x - a| : x \in A\}$, which is nonempty and bounded below. Prove that a nonempty set $A \subseteq \mathbb{R}$ is closed if and only if every number that has distance 0 from A is in A . (*Tutorial 5, Additional Question 1; Lecture 14, ε -Criterion for Closures; Lecture 1–3, ε -Criterion for Infima; builds on Question 3.*)

Solution

Method 1: ε -Criterion for Infima and for Closures

Proof. Write $d(a, A) := \inf\{|x - a| : x \in A\}$; 0 is a lower bound of this set.

$$\begin{aligned} d(a, A) = 0 &\iff \text{for every } \varepsilon > 0 \text{ there is } x \in A \text{ with } |x - a| < \varepsilon && (\varepsilon\text{-Criterion, inf}) \\ &\iff a \in \overline{A} && (\varepsilon\text{-Criterion, closure}) \end{aligned}$$

$(\Rightarrow)$ A closed $\implies \overline{A} = A$ (Closure Criterion for Closed Sets). $d(a, A) = 0 \implies a \in \overline{A} = A$.

$(\Leftarrow)$ Let $a \in \overline{A}$. Then $d(a, A) = 0$, so $a \in A$ by hypothesis. $\therefore \overline{A} \subseteq A \subseteq \overline{A}$, so $\overline{A} = A$ and A is closed (Closure Criterion for Closed Sets). $\square$

$$A \text{ closed} \iff (d(a, A) = 0 \implies a \in A) \text{ for every } a \in \mathbb{R}.$$

Question A2 (Disjoint closed sets can have distance 0)

Problem

For two sets $A, B \subseteq \mathbb{R}$, we define the *distance* between them as the infimum of the set $\{|a - b| : a \in A, b \in B\}$. If two closed sets A and B are disjoint (i.e., $A \cap B = \emptyset$), must the distance between them necessarily be positive? (*Tutorial 5, Additional Question 2; Lecture 13–15, closed sets and distance to a set; contrasts with Additional Question 3, the compact case.*)

Solution

Method 1: explicit counterexample

Lemma. If $\delta > 0$ and $S \subseteq \mathbb{R}$ satisfies $|s - s'| \geq \delta$ for all distinct $s, s' \in S$, then S is closed.

Proof of Lemma. Let $x = \lim x_k$ with every $x_k \in S$. There is K with $|x_k - x| < \frac{\delta}{2}$ for all $k \geq K$. For $k, l \geq K$: $|x_k - x_l| < \delta \therefore x_k = x_l$. So $\{x_k\}$ is eventually constant, equal to some $c \in S$. $\therefore x = c \in S$, and S is closed (Sequential Criterion for Closed Sets). $\square$

Proof. Let $A := \mathbb{N}$ and $B := \{n + \frac{1}{n} : n \in \mathbb{N}, n \geq 2\}$.

$A \cap B = \emptyset$: $n < n + \frac{1}{n} < n + 1$ for $n \geq 2 \therefore$ no element of B is an integer.

A closed: distinct integers differ by ≥ 1 (Lemma, $\delta = 1$).

B closed: distinct elements are $m + \frac{1}{m}, n + \frac{1}{n}$ with WLOG $m > n \geq 2$, and

$$\left(m + \frac{1}{m}\right) - \left(n + \frac{1}{n}\right) = (m - n) - \left(\frac{1}{n} - \frac{1}{m}\right) > 1 - \frac{1}{2} = \frac{1}{2}$$

$\therefore$ Lemma with $\delta = \frac{1}{2}$.

Distance 0: $|n - (n + \frac{1}{n})| = \frac{1}{n}$. Let $\varepsilon > 0$; choose $n \geq 2$ with $n > \frac{1}{\varepsilon}$ (Archimedean) $\therefore \frac{1}{n} < \varepsilon$. 0 is a lower bound, so $\inf\{|a - b| : a \in A, b \in B\} = 0$ (ε -Criterion for Infima). $\square$

No: $A = \mathbb{N}$ and $B = \{n + 1/n : n \geq 2\}$ are disjoint, closed, and at distance 0.

Question A3 (Disjoint compact sets have positive, attained distance)

Problem

Recall that for two sets $A, B \subseteq \mathbb{R}$, we define the *distance* between them as the infimum of the set $\{|a - b| : a \in A, b \in B\}$.

Prove that if two (sequentially) compact sets K_1 and K_2 are disjoint, then the distance between them is positive and is attained by two numbers $a_1 \in K_1$ and $a_2 \in K_2$; i.e., if the sets K_1 and K_2 are (sequentially) compact and have an empty intersection, then there are two numbers $a_1 \in K_1$ and $a_2 \in K_2$ for which the distance between K_1 and K_2 is $|a_1 - a_2|$, which is positive. (*Tutorial 5, Additional Question 3; Lecture 16, Sequential Compactness and Extrema of Closed and Bounded Sets; Lecture 4–6, triangle inequality in the limit sense; contrasts with Additional Question 2.*)

Solution

Method 1: compactness of the distance set; Extrema of Closed and Bounded Sets

Proof. Let K_1, K_2 be nonempty, compact, disjoint, and $S := \{|a - b| : a \in K_1, b \in K_2\} \neq \emptyset$.

S is compact. Let $x_n = |a_n - b_n| \in S$ with $a_n \in K_1, b_n \in K_2$. K_1 compact: $a = \lim a_{n_k}$ for some $a \in K_1$. K_2 compact: $b = \lim b_{n_k}$ for some $b \in K_2$; also $a = \lim a_{n_{k_j}}$ (Subsequential Limits). Put $u_j := a_{n_{k_j}} - b_{n_{k_j}}$ and $u := a - b$; then $u = \lim u_j$ (ALT).

$$||u_j| - |u|| \leq |u_j - u| \quad (\text{reverse triangle inequality})$$

$\therefore |u| = \lim |u_j| = \lim x_{n_{k_j}}$ (Squeeze), and $|u| = |a - b| \in S$. So S is compact.

$0 \notin S$. $|a - b| = 0 \implies a = b \in K_1 \cap K_2 = \emptyset$, impossible. $\therefore S \subseteq (0, \infty)$.

Conclusion. S is closed and bounded (Sequential Compactness of Closed and Bounded Sets) and nonempty. $\therefore m := \min S$ exists (Extrema of Closed and Bounded Sets), so $m = |a_1 - a_2|$ for some $a_1 \in K_1, a_2 \in K_2$. $m \in S \subseteq (0, \infty) \therefore m > 0$; $m = \min S = \inf S = \text{dist}(K_1, K_2)$. $\square$

$\text{dist}(K_1, K_2) = |a_1 - a_2| = \min S > 0 \text{ for some } a_1 \in K_1, a_2 \in K_2.$

Method 2: minimising sequence

Proof. $d := \inf S$ exists ($S \neq \emptyset$, 0 is a lower bound). Choose $a_n \in K_1, b_n \in K_2$ with $d \leq |a_n - b_n| < d + \frac{1}{n}$ (ε -Criterion for Infima). Extract twice as above: $a = \lim a_{n_j} \in K_1$ and $b = \lim b_{n_j} \in K_2$. Then $|a - b| = \lim |a_{n_j} - b_{n_j}|$ (as above). $d \leq |a_{n_j} - b_{n_j}| < d + \frac{1}{n_j}$ with $n_j \geq j \therefore d = \lim |a_{n_j} - b_{n_j}| = |a - b|$ (Squeeze). $K_1 \cap K_2 = \emptyset \implies a \neq b \therefore d = |a - b| > 0$ is attained at $a \in K_1, b \in K_2$. $\square$

Method Map

Q.	Core topic	Methods included
1	Limit points / isolated points / closure of finite sets	Sequential Criterion for Limit Points (injective-sequence lemma); density of $\mathbb{Q}$ (Archimedean); telescoping bound + MCT; Cauchy Criterion.
2	(Sequentially) compact or not	Closed and bounded; witnessing sequences.
3	$\sup A \in \overline{A}$	ε -Criterion for Suprema and for Closures.
4	Closed sets under union / arbitrary intersection	Sequential Criterion for Closed Sets; pigeonhole on indices.
5	Closed subset of a compact set is compact	Definition of compactness.
A1	Closed sets via distance-zero points	ε -Criterion for Infima and for Closures.
A2	Disjoint closed sets, distance 0	Explicit counterexample; uniform-gap lemma.
A3	Disjoint compact sets, positive attained distance	M1: compactness of $\{ a - b \}$, Extrema of Closed and Bounded Sets. M2: minimising sequence.

Final Answers

1. A : limit points $\{-1, 1\}$, isolated $A \setminus \{1\}$, closure $A \cup \{-1\}$. B : $[0, 1]$, $\emptyset$, $[0, 1]$. C : $\mathbb{R}$, $\emptyset$, $\mathbb{R}$. D : $\{s\}$ ($s = \sum_{n=1}^{\infty} 1/n^2$), D , $D \cup \{s\}$. E : $\emptyset$, E , E .
 2. (a) Compact. (b) Not compact. (c) Not compact. (d) Not compact. (e) Not compact.
 3. $\sup A \in \overline{A}$ for every nonempty, bounded-above $A \subseteq \mathbb{R}$.
 4. (a) F_1, F_2 closed $\implies F_1 \cup F_2$ closed. (b) An arbitrary intersection of closed sets is closed.
 5. A closed subset of a compact set is compact.
- A1.** A is closed $\iff$ every number at distance 0 from A lies in A .
- A2.** No: $A = \mathbb{N}$ and $B = \{n + 1/n : n \geq 2\}$ are disjoint, closed, at distance 0.
- A3.** $\text{dist}(K_1, K_2) = |a_1 - a_2| > 0$ for some $a_1 \in K_1, a_2 \in K_2$.